

Generation Interconnection System Impact Study Report

For Queue Project Q13-0296

Chesterfield Ave 138 kV

November 30, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0296. The Interconnection Customer proposes a 785.9 MW natural gas combined cycle facility interconnecting at the Chesterfield Ave 138 kV substation in the West Virginia area. The request was received on December 02, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 155 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$65,184,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Garces – Cap Sandario 138.0 kV is overloaded to 232.1% of its 223 MVA emergency rating for the loss of the Capitol Hill Substation – Chesterfield Ave 138.0 kV. The project contributes 70.6% DFAX (555.1 MW) to this overload.

Cap Sandario Substation – Chesterfield Ave 138.0 kV: 212.1% loading (247 MVA) following the outage of the Chesterfield Ave – Capitol Hill 138.0 kV; DFAX = 70.6%.

For the loss of Chesterfield Ave – Cap Sandario 138.0 kV L, the Capitol Hill – Chesterfield Ave 138.0 kV loads to 193.8% of its 237 MVA rating. Project DFAX: 65.1%.

Washington St – Capitol Hill 138.0 kV: 185.3% loading (244 MVA) following the outage of the Chesterfield Ave – Cap Sandario 138.0 kV; DFAX = 65.1%.

For the loss of Cap Sandario – Chesterfield Ave 138.0 kV, the Washington St – Chemical 138.0 kV loads to 172.9% of its 256 MVA rating. Project DFAX: 65.1%.

For the loss of Capitol Hill – Chesterfield Ave 138.0 kV, the Brownfield Switching – Garces 138.0 kV loads to 145.9% of its 229 MVA rating. Project DFAX: 48.0%.

Eagle Vly In – Brownfield Switching 138.0 kV line: 137.4% loading (236 MVA) following the outage of the Capitol Hill – Chesterfield Ave 138.0 kV; DFAX = 48.0%.

Conrad – South Ridge 138.0 kV: 136.6% loading (153 MVA) following the outage of the Cap Sandario – Chesterfield Ave 138.0 kV L; DFAX = 24.5%.

Chesterfield Ave – Lodi Stig 138.0 kV line: 134.6% loading (233 MVA) following the outage of the Cap Sandario – Chesterfield Ave 138.0 kV Line; DFAX = 34.9%.

The Clendnin – Garces 138.0 kV is overloaded to 122.7% of its 180 MVA emergency rating for the loss of the Garces – Brownfield Switching 138.0 kV Line. The project contributes 27.7% DFAX (217.5 MW) to this overload.

For the loss of International Paper – Flatlick 500.0 kV, the Myrtle-Date – Waterford Power Llc 230.0 kV loads to 112.1% of its 494 MVA rating. Project DFAX: 7.3%.

The Clinton Atle – Farragut 138.0 kV is overloaded to 111.5% of its 159 MVA emergency rating for the loss of the Cap Sandario – Chesterfield Ave 138.0 kV line. The project contributes 22.3% DFAX (174.9 MW) to this overload.

S Ridge – Silverado 138.0 kV L: 108.4% loading (203 MVA) following the outage of the Chesterfield Ave – Cap Sandario 138.0 kV; DFAX = 24.5%.

Silverado – Chemical Substation 138.0 kV Line: 104.2% loading (223 MVA) following the outage of the Chesterfield Ave – Cap Sandario 138.0 kV; DFAX = 24.5%.

The Clymer – Conrad 138.0 kV L is overloaded to 100.9% of its 198 MVA emergency rating for the loss of the Cap Sandario Substation – Chesterfield Ave 138.0 kV Line. The project contributes 24.5% DFAX (192.9 MW) to this overload.

For the loss of Fishkill Plains – DillardOrange 345.0 kV, the Solana – Northern Steel Casting 138.0 kV loads to 100.2% of its 239 MVA rating. Project DFAX: 5.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Garces – Cap Sandario 138.0 kV	Rebuild line with higher-capacity conductor	\$1,024,000	100% (\$1,024,000)
Cap Sandario Substation – Chesterfield Ave 138.0 kV	Rebuild line with higher-capacity conductor	\$2,123,000	100% (\$2,123,000)
Capitol Hill – Chesterfield Ave 138.0 kV	Rebuild line with higher-capacity conductor	\$1,397,000	96% (\$1,347,000)
Washington St – Capitol Hill 138.0 kV	Rebuild line with higher-capacity conductor	\$3,441,000	96% (\$3,317,000)
Washington St – Chemical 138.0 kV	Rebuild line with higher-capacity conductor	\$2,405,000	96% (\$2,318,000)
Brownfield Switching – Garces 138.0 kV	Rebuild line with higher-capacity conductor	\$2,369,000	100% (\$2,369,000)
Brownfield Switching – Eagle Vly In 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,910,000	100% (\$4,910,000)
Conrad – South Ridge 138.0 kV	Rebuild line with higher-capacity conductor	\$6,686,000	87% (\$5,833,000)
Chesterfield Ave – Lodi Stig 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,623,000	94% (\$3,388,000)
Clendenin – Garces 138.0 kV line	Rebuild line with higher-capacity conductor	\$26,083,000	92% (\$23,914,000)
Myrtle-Date – Waterford Power Llc 230.0 kV	Rebuild line with higher-capacity conductor	\$26,131,000	11% (\$2,972,000)
Farragut Substation – Clinton Atle 138.0 kV	Rebuild line with higher-capacity conductor	\$7,184,000	93% (\$6,662,000)
South Ridge Substation – Silverado 138.0 kV	Reconductor line	\$536,000	87% (\$467,000)
Chemical – Silverado 138.0 kV	Reconductor line	\$1,731,000	87% (\$1,510,000)
Clymer – Conrad 138.0 kV L	Reconductor line	\$869,000	87% (\$758,000)
Solana – Northern Steel Casting 138.0 kV	Reconductor line	\$2,728,000	83% (\$2,272,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$65,184,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.